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Electron emission physics simulations

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This is the final technical report for the project. It has the following structure: We begin by introducing the personnel employed using the AFOSR grant money, as well as others that were not funded by the grant but took part in project activities; subsequently we describe research activity and accomplishments in relation to the initial objectives of the project proposal; we then describe the impact of the project. Finally, we will give a financial report and discuss changes to the project.
The work done in this project is centered around use of the Reykjavík University Code for Molecular Dynamics Modeling of Electron Emission and Dynamics (RUMDEED), which is simulation software that has been developed at Reykjavík University over the years. Thus, the project has benefited both direct scientific research and further development of the software.

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FINAL REPORT

Award no. FA-8655-23-1-7003

Electron Emission Physics Simulations.

PI: Ágúst Valfells

March 1, 2023 – February 28, 2025

Contents

Introduction

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Personell and collaboration

Personnel

Over the course of this project the following have been employed:

Kristinn Torfason (KT), research Scientist

- 40% employment from March 2023 – May 2024, and December 2024 – February 2025.
- 100% employment from June 2024 – November 2024.
- Has worked on all aspects of the project.
- Software development and oversight.

Hákon Örn Árnason (HÖÁ), doctoral student

- Employed for 1 man-month 2023.
- Generation of simulation results for photoemission from heterogeneous cathode.
- Data curation.

Þorsteinn Hanning Kristinsson (ÞHK), master's student

- Employed for 4.5 man-months 2023.
- Generation of simulation results for field emission.
- Data curation.

Arnar Jónsson (AJ), undergraduate student

- Employed for 3 man-months in 2023, and 4 man-months in 2024-2025.
- Generation of simulation data for space-charge limited emission.
- Generation of simulation data for field emission.
- Recombination and ionization modeling.
- Implementation of dedicated finite-difference solver in RUMDEED (unpaid coursework).
- Data curation.

Bjartþór Steinn Alexandersson (BSA), undergraduate student

- Employed for 3 man-months in 2024-2025.
- Generation of simulation data for beam loading.
- Data curation.

The PhD student Yuan Zhou, who has been supported by the RU doctoral fund, has been working with Kristinn Torfason on image-charge and reduced parameter modeling full time.

Note that, due to how the work on ultrafast emission has developed, it was decided to postpone work on emission from core-shell nanowires until work on strong-field emission from a conventional cathode was more mature. For that reason, Dr. Anna Maria Sitek has not been working on the project, and the contribution from Kristinn Torfason and the student research assistants has been increased.

Collaboration

AFRL: Ágúst Valfells (AV) and Kristinn Torfason (KT) made use of a Windows on Science grant to visit AFRL in Albuquerque in the summer of 2023. During his sabbatical year in 2024, AV also visited AFRL separately in the spring and fall. During these visits we had discussions with Dr. Travis Garrett, Dr. Rene Van Ginhoven, and Dr. Jeanne M. Riga on electron emission simulation in general, emission from carbon nanotube fibers, and ultrafast emission. In particular, the topic of distribution of momentum and tunneling time for electrons released via strong-field emission is a topic of ongoing interest and discussion.

Purdue University: Ágúst visited Prof. Allen Garner's group at Purdue University in the spring and fall of 2024. The Purdue group has been using the RUMDEED code with technical support from KT. Topics that we have been working upon are discrete particle effects on emission, and benchmarking of nexus theory for emission, both of which relate directly to this project. As part of the latter project, AV is serving on the master's committee of Purdue student, Allison Komrska. In addition, we have worked on capacitive models for space-charge limited emission on crossed field diodes (especially identifying their limitations) and on electrodynamics in reduced dimensions.

University of Bucharest: We have been in collaboration with Prof. George Nemnes at the University of Bucharest. His students, Calin Pantis and Amanda Preda, have visited Reykjavík University and worked with us on the use of deep neural networks for modelling of space-charge effects on general emitter geometries. Using Erasmus funds, AV visited the University of Bucharest in February of 2024 and gave a talk on the RUMDEED code and discrete electron effects. He also used the opportunity to visit the Extreme Light Infrastructure in Bucharest.

IVNC related: In April 2024, AV and KT attended the International Vacuum Electronics Conference in Monterey. At this time, we inquired whether there might be interest in having an International Vacuum Nanoelectronics Conference in Iceland in the future. Due to the intended host of the 2025 IVNC conference dropping out, we were asked on short notice to submit a bid for hosting IVNC 2025. A bid was prepared during the summer of 2024 and proved successful. This has opened new opportunities for collaboration as a sizeable number of scientists in the field will visit Reykjavík University for the conference in July 2025. We have already begun to forge new ties as AV visited Prof. Jan Dziuban's group at Wroclaw University of Technology in Poland. The Wroclaw group has considerable capabilities and experience in designing and manufacturing vacuum electronics-based MEMS devices, and we have identified some areas of mutual interest. This visit was funded by an Erasmus grant. In addition, we hope to leverage the conference to establish ties with groups doing experimental work on strong-field emission at the Friedrich-Alexander Universität in Erlangen and simulations at the Paul Scherrer Institute.

Other: AV visited the University of Maryland in the spring of 2024. There he met Dr. Kevin Jensen, among others. Dr. Jensen has been working on emission from rapidly time-varying barriers with Dr. Riga of AFRL. There is an interest in continuing discussions on this topic and incorporate the results of their work on the distribution of tunneling time and momentum of electrons into the RUMDEED model for ultrafast emission.

Accomplishments.

Reduced parameter modelling

The effort here has been on generating data from MD simulations using the RUMDEED code to be used as a basis for reduced order models. For this purpose, three students have been employed to run simulations. Hákon Örn Árnason, a Ph.D. student, was hired for one month to run simulations on photoemission from a heterogeneous cathode. Þorsteinn Hanning Kristinsson, was hired to work on field emission simulations, and Arnar Jónsson an undergraduate student was hired to run simulations on space-charge limited emission and field emission for different diode configurations. The main interest is to be able to generate I-V curves and to see how emittance is affected by parameters such as diode dimension, applied potential, cathode work function and heterogeneity, etc. The work has not been published yet, but we anticipate submitting initial results for publication later in 2025. This work was overseen by AV and KT.

Image charge approximation

A specific task, related to the reduced order modelling, was to develop a computationally efficient and sufficiently accurate method to account for the induced surface-charge in geometries where an analytical image charge solution does not exist. In addition to KT, this problem was worked on by the Ph.D. student, Yuan Zhou, who is funded separately from this project, and the University of Bucharest students, Cantis Palin and Amanda Preda, who developed a deep neural network model in collaboration with Yuan Zhou. In short, it was found that simple, physically informed, image-charge models performed as well as the DNN based models, with the additional benefit that they conformed better to the RUMDEED code. For simulating space-charge effects on emission from a *straight CNT fiber*, we developed a model consisting of a conducting capped cylinder and image charges embedded within it (five image charges for each external electron). The exact location and magnitude of the five image charge partners is prescribed by the geometry of the capped cylinder and location of the external charge. A similar model was constructed for space-charge effects from a *looped CNT fiber*. This model consists of a single image charge partner for each external charge. The location and magnitude of the image charge is prescribed by the shape of the looped emitter and position of the external charge [1,2]. Additionally, Arnar Jónsson (AJ) has implemented a dedicated and efficient finite-difference solver into RUMDEED to calculate the space-charge effects from discrete electrons. The purpose of this is to make a direct comparison with the image-charge approach in RUMDEED. It also offers the possibility of a hybrid computational approach that uses a combined image charge and Laplacian approach that could prove to be advantageous in certain terms of computational efficiency and accuracy. This work by AJ was done as a capstone project in his undergraduate studies, and he will work on the hybrid approach during his upcoming master’s studies at RU.

Ultrafast emission

We address this as a semi-electrostatic problem where we initially use the same approach for field emission as is currently built into RUMDEED for a slowly varying surface barrier. Our method is to calculate the supply function of electrons independently and subsequently use a Monte-Carlo approach to determine whether an electron tunnels through the barrier. The supply function and escape probability are both dependent on the local field at the cathode, which in turn can be strongly influenced by space-charge. For our initial simulations we place the emitted electrons above the cathode at an elevation just beyond the location of the maximum of the potential barrier for emission and assign it a small initial velocity, which is necessary for the electron to propagate away from the cathode in finite time. To do this properly it is important to be able to dynamically estimate the characteristic length scale for the barrier. From theoretical considerations we have identified a useful length scale which describes the location of the potential maximum, distance between emitted particles, and elevation above the cathode that an emitted electron must reach before successive emission takes place [3]. This length scale is $\xi_* \equiv \left(\frac{q}{2\pi\varepsilon_0 E_0}\right)^{1/2}$, where E_0 is the electric field at the cathode surface in absence of space-charge. We see that the surface barrier maximum for emitted electrons lies between $0.35\xi_*$ - $0.40\xi_*$ above the cathode surface for a very wide range of configurations, that a lower bound of $1.4\xi_*$ exists for the spacing between adjacently emitted electrons, and that for a point emitter an electron must have travelled a distance of at least ξ_* before another one is subsequently emitted. Figure 1 depicts the spacing between neighboring electrons at the cathode as a function of the applied electric field strength under space-charge limited conditions. The theoretical lower bound for spacing, $1.4\xi_*$, is shown in red.

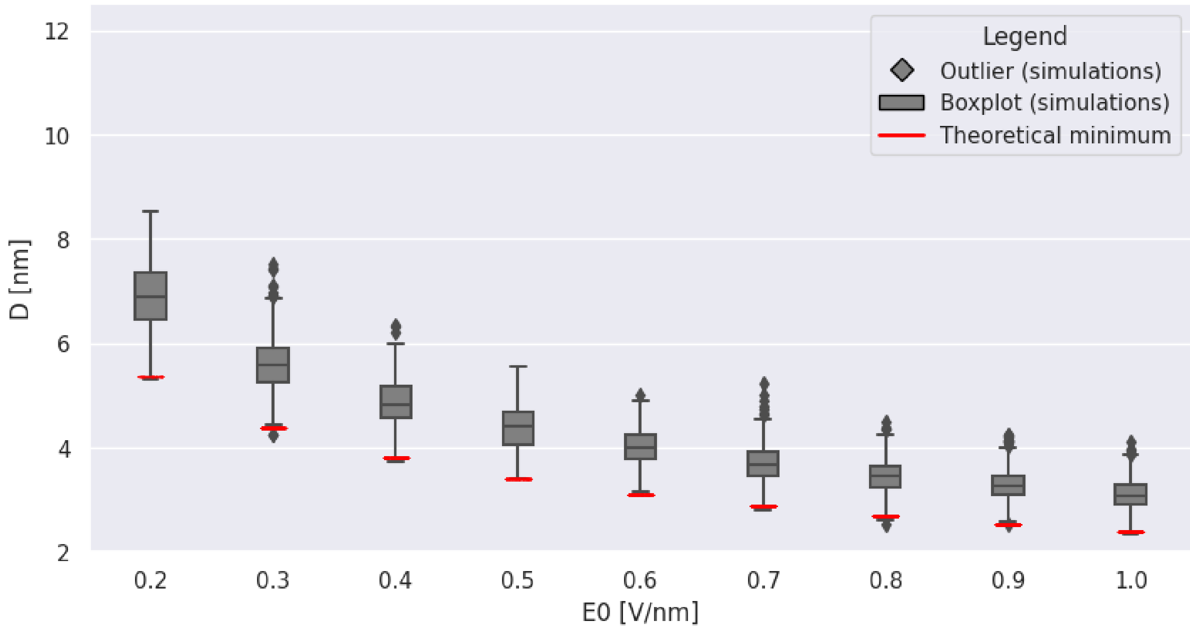

Figure 1. Spacing between neighboring electrons at the cathode surface as a function of applied field strength.

We can also estimate median values between emission sites as a function of the equilibrium surface electric field on the cathode in field emission. Although this work was motivated by modeling considerations for ultrafast emission, we find that it may be applied to find scaling laws for limiting current from various emitter geometries. As such it is also useful for general parametric models. Returning to the specific problem of ultrafast emission, we have begun simulations using a rapidly varying applied electric field in our semi-electrostatic model using our simple model of initial placement of electron. To calibrate our results, we compare emission in the absence of space-charge with theoretical models independent of our approach [4,5]. Subsequently we hope to incorporate the work being done by Jensen and Riga [6] to incorporate appropriate tunneling times and momenta to the emitted electrons.

Beam loading in fast switches

The undergraduate student, Bjartþór Steinn Alexandersson (BSA) has been assigned the task of simulating of how an external circuit connected to the diode affects the transient and steady state I-V characteristics of the diode current. Initial results, using only a resistive load, have been able to recover known results for equilibrium current in a field emitting diode with a base resistor [7], but they also show previously unknown effects on the temporal response, e.g. diminished overshoot and a longer time to reach equilibrium when the external resistance is comparable to the diode impedance in magnitude. BAS has been hired for the summer of 2025, using RU funds, to continue working on this topic.

Ionization and recombination

A model for inelastic collisions of electrons with molecular Nitrogen, including ionization, has been implemented and tested. Scattering from and ionization of a background gas is modelled using conventional Monte Carlo methods whereby the probability of interaction between an electron and gas molecule over a given time step are determined from the particle velocity and interaction cross sections. In the case of ionization, we assume that the ion is at rest and momentum is carried off by the two electrons. For ionization we assume that the background gas density is prescribed and constant. When implementing a model for electron-ion recombination, it was found that such a continuity assumption was not appropriate, and a discrete recombination model was developed. We have also developed a discrete model for collisions and ionization, which may be more appropriate than the continuous model in the case where the mean free path for collisions is long compared to system size. For recombination we assume that radiative recombination is the dominant mechanism. To model this, we use the vectorized positions of all electrons and ions in the diode gap to determine if any of the electrons are within a sphere of radius $R_r(Z, v)$ from any of the ions. If this is the case, recombination occurs. This radius is derived from a theoretical cross-section for recombinant radiation due to Kramers [8] that depends on the electron velocity, v, and effective charge of the ion, Z [9]. Even though recombination is comparatively rare compared to the transit time of electrons across the gap, we find that it effectively leads to a stochastic equilibrium in the number of ions in the diode gap [10]. Figure 2 shows how the number of ions evolves over time in a long simulation. The time to equilibrium is roughly 300 times as long as the transit time for a single electron across the diode gap. The green curve shows the development of ion number if, initially, 60% of the background nitrogen gas was ionized; the red curve shows the results if the initial ionization was 20%; and the blue curve shows the results if the background gas was initially not ionized at all.

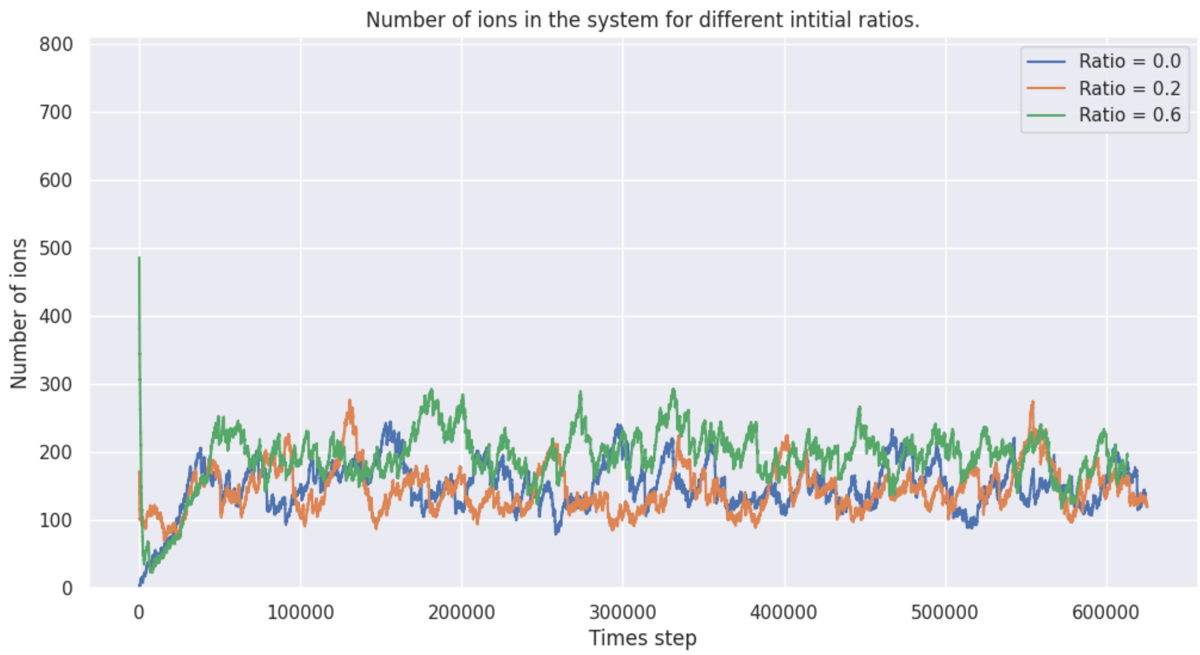

Figure 2. Evolution of ion number in the diode gap because of electron ionization of molecular Nitrogen and radiative recombination. The different curves represent different fractions of background gas that was ionized at the start of the simulation: 60% (green), 20% (red), and 0% (blue).

Dielectronic recombination may be added in the future. Two-time-scale dynamics have been implemented in the RUMDEED code, though extensive simulations remain to be done. We find that the appropriate, longer time-step for ions can be calculated from the I-V response of the electrons. This is characteristically twice the transit time for a single electron across the diode gap. Thus the two-time-step model can run much faster than a one-time-step model for long simulations that are meant to include ion dynamics.

Ion enhanced field emission

We are commencing detailed simulations of these effects, now that the recombination and two-time-scale features have been implemented in RUMDEED.

Ultrafast emission from nanowires

This work has been postponed until the results on tunneling times and initial momenta have matured.

Impact

Personnel

Four students have been employed to work on this project to various degrees. This type of research experience has in the past proven to be especially valuable for students who are not yet pursuing a research degree. It has opened their eyes to the joy of discovery and opened the doors of excellent graduate programs outside of Iceland.

Hákon Örn Árnason graduated with a Ph.D. from RU in June 2023. He is currently the Tactical Data Link Manager for the Integrated Air Defense Systems Engineering Department of the Icelandic Coast Guard.

Þorsteinn Hanning Kristinsson is currently a doctoral student in Aerospace Engineering at the University of Colorado, where he is conducting research on passive radar applications for planetary missions.

Arnar Jónsson will graduate with a BSc degree in Engineering from RU in June 2025 and will commence master's studies at RU in the fall. He will continue working on material related to this project. He is an exceptional student and plans on pursuing a Ph.D. in Computational Engineering abroad.

Bjartþór Steinn Alexandersson will continue his undergraduate studies in Engineering at RU. He will be funded by RU to work on simulating beam loading this summer. He is also a first-rate student that is likely to remain with our research group for the remainder of his undergraduate studies.

Publications and presentations

In this section we give an overview of presentations at scientific conferences, workshops and seminars, and proceedings papers. We have not yet submitted a full-length scientific article on the work funded by this grant, but anticipate submitting, by the end of the year: three articles to the *Journal of Vacuum Science and Technology B* in connection with papers presented at IVNC 2025; and one paper on the RUMDEED code to *Physics of Plasmas*. We already have the material ready for these articles.

Published in conference proceedings:

Y. Zhou, K. Torfason, C. Pantis, A. Manolescu, Á. Valfells, 'Machine learning methods for computing discrete image charge effects of a nanowire electron emitter', *2024 IEEE International Conference on Plasma Science*, 16-20 June 2024, DOI: 10.1109/ICOPS58192.2024*2024*.

Accepted for conference proceedings:

A. Jónsson, K. Torfason, A. Manolescu, Á. Valfells, 'Molecular dynamics simulation of discrete particle ionization and recombination in a thin gas in a vacuum nanodiode', *IEEE Pulsed Power & Plasma Science Conference*, 15-20 June 2025, Berlin.

Y. Zhou, Á. Valfells, K. Torfason, A. Manolescu, 'Image charge approximations for CNT-fiber emitters in a molecular dynamics code', *IEEE Pulsed Power & Plasma Science Conference*, 15-20 June 2025, Berlin.

K. Torfason, A. Manolescu, Y. Zhou, A. Jónsson, Á. Valfells, 'RUMDEED: A molecular dynamics code for simulating nano- and microscale vacuum electronics', *IEEE Pulsed Power & Plasma Science Conference*, 15-20 June 2025, Berlin.

Á. Valfells, K. Torfason, A. Jónsson, A. Manolescu, 'Discrete particle effects on electron emission', *IEEE Pulsed Power & Plasma Science Conference*, 15-20 June 2025, Berlin.

A. Jónsson, K. Torfason, A. Manolescu, N. Pop, Á. Valfells, 'Molecular dynamics modelling of ionization and recombination in a vacuum diode', *IEEE International Vacuum Nanoelectronics Conference 2025*, 8-11 July 2025, Reykjavík.

Y. Zhou, K. Torfason, A. Manolescu, Á. Valfells, 'Adaption of image charge approximations to general emitter shapes', *IEEE International Vacuum Nanoelectronics Conference 2025*, 8-11 July 2025, Reykjavík.

Á. Valfells, K. Torfason, A. Jónsson, A. Manolescu, 'Granularity in electron emission at the nanoscale', *IEEE International Vacuum Nanoelectronics Conference 2025*, 8-11 July 2025, Reykjavík.

Scientific talks:

K. Torfason, 'Tutorial: Simulating Nano- and Microscale Vacuum Electronics with Molecular Dynamics', *IEEE International Vacuum Nanoelectronics Conference*, 8-11 July 2025, Reykjavík.

Á. Valfells, 'Electron Emission and Propagation in the Immediate Vicinity of the Cathode', AFRL, Albuquerque, May 2023.

Á. Valfells, 'Electron Emission and Propagation in the Immediate Vicinity of the Cathode', University of Bucharest Dept. of Physics, March 2024.

Á. Valfells, 'Microscale Effects on Electron Emission and Propagation', Purdue University Dept. of Nuclear Engineering, May 2024.

Á. Valfells, 'Discrete Charge and Electron Emission', Purdue University Dept. of Nuclear Engineering, October 2024.

Software

The capabilities of the RUMDEED code have been significantly enhanced during this project. A list of major improvements follows.

Space-charge effects for more general geometry:

- Simplified image charge models for capped cylinders and looped cylinders have been constructed and implemented in RUMDEED.
- A dedicated Laplace solver for standard geometries has been implemented into RUMDEED.
- Preparations for a hybrid image charge / Laplace solver have begun.
- Deep neural network models for induced surface charge on capped cylinders were developed but not implemented in RUMDEED as they were not a significant improvement over the simplified image charge models.

Ion effects:

- Implemented ionization models in RUMDEED for molecular Nitrogen. One assumes continuous background gas, the other is based on discrete gas molecules.
- Implemented model for radiative recombination in RUMDEED. Model works with discreet ions.
- Implemented two-time-step model for simulating electron and ion motion in a computationally efficient manner.

Ultrafast emission and discrete particle effects:

- Developed rudimentary model for adaptive determination of initial placement of emitted electrons in strong-field emission.

Parametric modelling:

Aside from the work on reduced order modelling of space-charge induced surface charge for generalized emitter shapes, data has also been generated to support construction of reduced order models or beam emittance, current and brightness.

- Generated dataset on I-V characteristics and emittance for photoemission.
- Generated dataset on I-V characteristics and emittance for photoemission.

- Generated dataset on I-V characteristics and emittance for enforced space-charge limited emission.

Other output

Building on this project we have twice submitted proposals to the Icelandic Research Fund for a Grant of Excellence. These amount to roughly one million dollars over three years. The first proposal, submitted in June 2023, received the second highest grade of A2 in evaluation and was not funded. An improved proposal was submitted in June 2024. It received the highest possible grade of A1, and we were invited to interview with the board of the Icelandic Research Fund. Unfortunately, funding for science has been reduced, and only 2 Grants of Excellence were awarded in this cycle as compared to 4 to 6 in previous cycles. Nonetheless, the reception was excellent, and the proposal will be submitted with changes due to advancements that have been made in this AFOSR supported project.

We also had the opportunity to organize and host the 2025 International Vacuum Electronics Conference. This will take place at RU in July of 2025. The support we have received from AFOSR in this project, and a previous one, has benefited the work on vacuum electronics at RU and greatly strengthened our capacity for hosting the conference.

Financial report

Table 1 shows the overview of cash flow throughout the reporting period in Icelandic króna (ISK). The exchange rate was roughly 140 ISK to the US dollar over the reporting period.

Period	Payments received	Payments made	Remarks
March 2023	9,866,500.00	1,299,889.00	$70,000 from funding agency KT, HÖÁ, ÞHS
April 2023	0	614,982.00	KT, ÞHK
May 2023	6,253,200.00	614,982.00	$45,000 from funding agency KT, ÞHK
June 2023		1,116,647.00	KT, ÞHK, AJ
July 2023		1,026,450.00	KT, ÞHK, AJ
August 2023		831,814.00	KT, AJ
September 2023		330,149.00	KT
October 2023		330,149.00	KT
November 2023	3,583,000.00	330,149.00	$25,000 from funding agency KT
December 2023		330,149.00	KT
January 2024		358,028.00 (330,149.00)	KT
February 2024		358,028.00 (330,149.00)	KT
March 2024		890,282.00	KT, AJ
April 2024		358,028.00	KT
May 2024		358,028.00	KT
June 2024		1,031,042.00	KT (100%)
July 2024		1,031,042.00	KT (100%)
August 2024		1,031,042.00	KT (100%)
September 2024		1,031,042.00	KT (100%)
October 2024		1,031,042.00	KT (100%)
November 2024		1,031,042.00	KT (100%)
December 2024		1,456,028.00	KT, AJ, BSA
January 2025		1,456,028.00	KT, AJ, BSA
February 2025		1,456,028.00	KT, AJ, BSA
All	19,702,700.00	19,702,090.00	

Table 1. Monthly overview of cash flow through reporting period (in ISK).

All costs charged to this grant have been for salaries of the researcher Kristinn Torfason and the student research assistants, Hákon Örn Árnason, Þorsteinn Hanning Kristinsson, Arnar Jónsson, and Bjartþór Steinn Alexandersson. Please note the Iceland króna (ISK) has been used as the currency of record. This is because salaries and taxes are fixed in ISK, which has a variable exchange rate with the US dollar. Roughly speaking the exchange rate was about 140 ISK to the dollar over the reporting period.

Changes

There was some slight deviation from the initial financial plan due to how the results of investigations turned out. Specifically, the role of Dr. Anna Sitek in the project changed as the ultrafast emission investigations deviated from direct use of Greens' function formalism to incorporating a spread of electron momenta emerging from a rapidly time varying barrier into a quasi-electrostatic model of time-varying field emission. This will implement results from the theoretical work of our collaborator, Dr. Jeanne M. Riga of AFRL, that are more suitable for the RUMDEED code. Therefore, the decision was made to postpone Dr. Sitek's work on ultrafast emission from core-shell nanowires and reallocate resources to Kristinn Torfason and student research assistants to do other work related to ultrafast emission, parametrization and effects of an external system on temporal response. The students also worked on code documentation.

Note that there was a slight increase in salary costs for January and February of 2024 compared to what was presented in the previous annual report, from 330,149.00 ISK to 358,028.00 ISK. This is due to RU instituting a salary rise that was retroactive to January 1, 2024.

Beyond the initial objectives of the project, a dedicated Laplace solver was built into the RUMDEED code. This was initially done to help direct comparison and calibration of the parametric image charge models. However, it has become apparent that this solver may be used in a hybrid approach of simulation. The idea is to use image charge modelling in the regions where scattering and discrete particle effects are most important and a Laplacian solver for calculating space-charge effects from and forces on electrons that are outside those regions. It is anticipated that this can speed up computation while maintaining accurate representation of important discrete charge effects.

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List of Symbols, Abbreviations and Acronyms

AFOSR	Air Force Office of Scientific Research
AFRL	Air Force Research Laboratory
CNT	Carbon Nanotubes
DNN	Deep Neural Networks
ISK	Icelandic *króna*
MD	Molecular Dynamics
RU	Reykjavík University
RUMDEED	Reykjavik University Molecular Dynamics code for Electron Emission and Dynamics
USD	U.S. Dollar